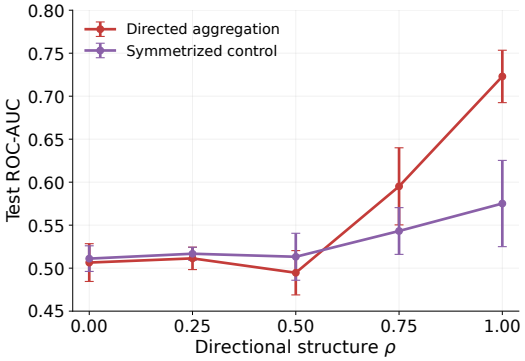
Edge orientation in link prediction

Idea: aggregate incoming and outgoing neighbors separately.



$\rho = 0$: random orientation. $\rho = 1$: cyclic roles.

Error bars: sample SD over 5 seeds.

Controlled comparison

Same skeleton, features and decoder.

Matched model size.

ROC-AUC at $\rho = 1$

0.723 vs **0.575**

Paired gain:

$\Delta = 0.148 \pm 0.020$ (SD).

No consistent gain at weak directional structure.

Synthetic graph: 180 vertices, three roles. Result applies to this generating family.

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